

Exercise 1

a) We need to prove that $\text{CNF-preSAT}(k)$ is in NP by:

- providing certificate y for an input φ , with $|y|$ polynomial in the size of φ
- giving a polynomial algorithm which checks the $\text{CNF-preSAT}(k)$ property for a given input φ and certificate y

We also need a proof that $\text{CNF-preSAT}(k)$ is NP-hard. For that we need to show that some other NP-hard problem reduces to $\text{CNF-preSAT}(k)$

b) $\text{CNF-preSAT}(k)$ is ~~an~~ NP:

certificate $\rightarrow v: \text{Atoms} \rightarrow \{0,1\}$, that satisfies all clauses of φ apart from at most k . We can check in polynomial time that v satisfies all clauses apart from at most k , by going through the clauses of φ and counting those which are not satisfied. Once all the clauses have been checked we compare this number to k .

CNF-preSAT(k) is NP-hard:

We show that $\text{CNF-SAT} \leq_p \text{CNF-preSAT}(k)$.

We want to convert any input of CNF-SAT into an input for CNF-preSAT(k) such that solving the second will give us the answer to the first.

We make a $f: \text{CNF} \rightarrow \text{CNF}$

$$f(\varphi) = \left(\bigwedge_{i=1}^k a_i \wedge \neg a_i \right) \wedge \varphi$$

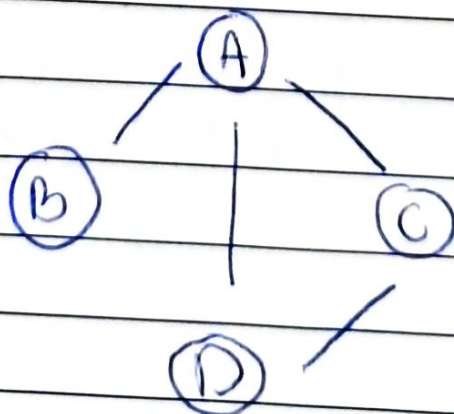
that way there is k unsatisfiable clauses and thus φ must be satisfiable.

$$\varphi \in \text{CNF-SAT} \iff f(\varphi) \in \text{CNF-preSAT}(k)$$

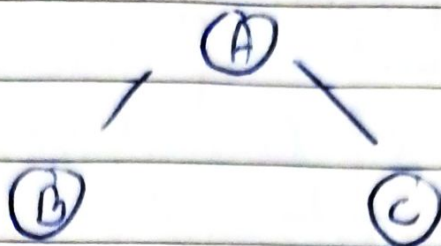
Exercise 2.

a)

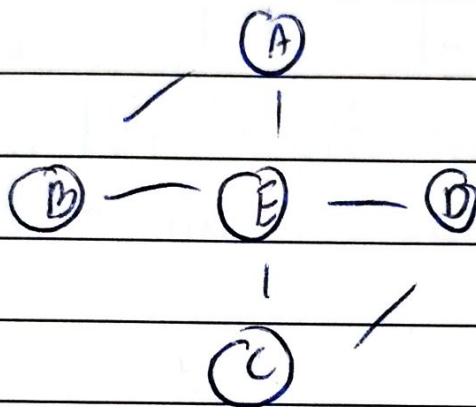
G :



G' :



Not isomorphic



b) This is NP:

The certificate is a unique mapping $f: V_1 \rightarrow V_2$. We check that for every edge (u, v) in G_1 , the corresponding edge $(f(u), f(v))$ exists in G_2 . This check is done in polynomial time.

It is NP-hard:

We show that ~~Clique(k) is a Problem~~ $\text{Clique}(k)$ is a Problem (G, k) we can reduce $\text{Clique}(k)$ to this problem. We take graph G and number k and we build a new graph full- k

Which is a complete graph on k nodes.
Now we check ~~with~~ if the new graph is a subgraph of G . If yes, then G must contain k nodes that form a clique since $\text{full-}k$ is a clique. If G has a clique of size k , then that clique is $\text{full-}k$ and is isomorphic to that subgraph.

Checking a $\text{full-}k$ if k is big is exponential so if k is bigger than $|V|$ in G it is impossible for G to contain a clique of size k .

Exercise 3.

a) Halt is NP-hard. To prove that we $\text{SAT} \leq_p \text{Halt}$. We define a Turing machine M that takes φ as input. It will try all possible combinations of true and ~~false~~ false assignments. If it finds an assignment that ~~so~~ satisfies the formula then it halts. If it's not satisfiable the machine will loop forever.

b) This problem is undecidable, so not NP, therefore not NP-complete.